

J2EE ARCHITECTURE

The diagram represents the **J2EE (Java 2 Enterprise Edition) Architecture**, which follows a **three-tier architecture model**. This architecture separates an application into three logical layers to improve scalability, security, and maintainability.

The three tiers are:

1. Client Tier
 2. Middle Tier (Application Server)
 3. Data Tier
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Client Tier

The **Client Tier** is the front-end layer where the user interacts with the application.

Examples:

- Web Browser (Chrome, Firefox)
- Desktop application
- Mobile application

Responsibilities:

- Sends requests to the server
- Receives responses from the server
- Displays the output to the user

In web applications, the browser acts as the client. It sends HTTP requests to the server and displays HTML responses.

Middle Tier (J2EE Server)

The **Middle Tier** is the core processing layer. It contains the J2EE server, which handles business logic and request processing.

This tier is divided into:

A) Web Container

The Web Container manages:

- Servlets
- JSP (Java Server Pages)

Functions:

- Receives HTTP requests from the client
- Executes Servlets or JSP files
- Generates dynamic web content
- Sends responses back to the client

Example:

When a user submits a login form, the Servlet processes the data and verifies it.

Apache Tomcat is an example of a Web Container.

B) EJB Container

EJB stands for **Enterprise Java Beans**.

This container manages:

- Business logic
- Transaction management
- Security
- Enterprise-level operations

It is mainly used in large-scale enterprise applications like banking systems.

C) Standard Services

The J2EE server also provides built-in services such as:

- Security management
- Transaction management
- Naming services (JNDI)
- Messaging services

These services reduce developer effort and improve application reliability.

Data Tier

The **Data Tier** contains the database system.

Examples:

- MySQL
- Oracle
- PostgreSQL

Responsibilities:

- Stores application data
- Retrieves data when requested
- Maintains data integrity

The middle tier communicates with the database to store or fetch information.

